WORKSHEET

Converting rates

Litres/minutes, tonnes/day and cents/kilograms are examples of 'unit rates' because the rates are 'per one unit'. When we travel 70 km in 1 hour we write this average rate as the unit rate '70 km/h'.

1 Gavin rides 25 km in 30 minutes.

a At the same rate, how far will he travel in 1 hour?

b Write Gavin's average speed in km/h.

2 Rhea works as a supermarket check-out operator. She can scan nine items in 30 seconds.

a How many items can Rhea scan in one minute?

b How many items can she scan in one hour?

c Express Rhea's scanning rate in items per hour.

3 The valve on Noel's water tank is leaking. He measured that the tank lost 22 mL in 15 seconds.

a How many mL does the tank lose per hour?

b Express the rate that the tank is losing water in mL/h.

4 Sound travels at 1100 km/h.

a Express 1100 km in metres.

b How many metres does sound travel in one minute?

c How many metres does sound travel in one second? Express your answer correct to one decimal place.

d Express the speed of sound in m/s.

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5 During a training boat row, Ross rowed 1200 m in $2\frac{1}{2}$ minutes.

a Express $2\frac{1}{2}$ minutes in seconds.

b On average, how far did Ross row per second?

c Express Ross's average speed in m/s.

6 The amount of coal remaining in a mine is decreasing at a rate of 15 million tonnes per year. The mine operates 24 hours per day, 7 days per week. Calculate the rate the coal in the mine is decreasing, correct to the nearest whole number, in:

a tonnes per day (t/day)

b tonnes per hour (t/h)

c tonnes per second (t/s)

7 Mya is driving at an average speed of 80 km/h. At this speed:

a How many metres does she drive per hour?

b How many metres does she drive per second?

c Express a speed of 80 km/h in m/s.

8 Melanie's car used 15.9 L of petrol to travel 300 km.

a How many litres of petrol did the car use to travel 100 km?

b Express Melanie's car's fuel consumption rate in L/100km.

9 Caroline jogs at a speed of 6.5 km/h. Express this speed in m/s.

10 Jessica's hens lay eggs at the rate of 5 eggs per day. Express this rate as dozens of eggs per year.

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Answers

1 **a** 50 km

b 59 km/h

2 **a** 18

b 1080

c 1080 items/h

3 **a** 5280

b 5280 mL/h

4 **a** 1100 000 m

b 18 333.3 m

c 305.6 m

d 305.6 m/s

5 **a** 150 s

b 8 m

c 8 m/s

6 **a** 41 096 t/d

b 1712 t/h

c 0.5 t/s

7 **a** 80 000 m

b 22

c 22 m/s

8 **a** 5.3 L

b 5.3 L/100 km

9 1.8 m/s

10 152 dozen/year

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